

Replication Report:

Emergence of Nematic Loop-Current Bond Order in Kagome Metals near van Hove Singularities

Friedlan & Kee, arXiv:2510.05234v2 (2026)

Textures-100 overnight replication (loop-current class)

July 19, 2026

Abstract

We replicate the analytic core of the effective patch model of Friedlan & Kee for nematic loop-current bond order (NLCBO) in AV_3Sb_5 kagome metals. We rebuild the 6×6 Bloch Hamiltonian $H(\mathbf{k})$ (their Eq. 4) from scratch, in the same hexagonal-BZ convention as the shared loop-current kagome kernel, and verify five machine-checkable claims: (C1) the numeric spectrum reproduces the analytic unperturbed eigenvalue formula Eq. (9) to machine precision and is degenerate at total phase $\Phi = 0, \pi$; (C2) the order-parameter classification by Φ and by time-reversal-symmetry-breaking (TRSB) content, with NLCBO the unique nematic config; (C3) the inverse-energy factors Eq. (12) satisfy $1/\Delta E_1 > 0$, $1/\Delta E_2 < 0$ at the paper's parameters; (C4) the second-order-in- λ mechanism, in which NLCBO alone develops an anomalous k_x -only band dispersion that is the most negative of the three $\Phi = \pi$ phases while LCBO+ wins at full band filling; and (C5) that λ is required—the three $\Phi = \pi$ phases are exactly degenerate at $\lambda = 0$ and split only at $O(\lambda^2)$. **All 5/5 claims reproduce.** Coverage of the paper's claims: partial (the analytic patch-model mechanism is fully reproduced; the full mean-field simulated-annealing phase diagram and the 9-band DFT tight-binding model are not).

1 Paper in one paragraph

The AV_3Sb_5 kagome metals develop a $2a \times 2a$ charge-bond order (CBO) from nesting of van Hove singularities (vHS) at the three M points. Complex-valued bond modulations give loop-current order (LCO, TRSB). The authors introduce an effective *patch* model with one p -type (vH1) and one m -type (vH2) vHS at each M point, giving a 6×6 Hamiltonian in the basis $\{\psi_{2A}, \psi_{2B}, \psi_{2C}, \psi_{1A}, \psi_{1B}, \psi_{1C}\}$:

$$H(\mathbf{k}) = \begin{pmatrix} \epsilon & s_2 \Delta_{AB}^* & s_2 \Delta_{CA} & \lambda k_1 & 0 & 0 \\ s_2 \Delta_{AB} & \epsilon & s_2 \Delta_{BC}^* & 0 & \lambda k_2 & 0 \\ s_2 \Delta_{CA}^* & s_2 \Delta_{BC} & \epsilon & 0 & 0 & \lambda k_3 \\ \lambda k_1 & 0 & 0 & -\epsilon & s_1 \Delta_{AB} & s_1 \Delta_{CA}^* \\ 0 & \lambda k_2 & 0 & s_1 \Delta_{AB}^* & -\epsilon & s_1 \Delta_{BC} \\ 0 & 0 & \lambda k_3 & s_1 \Delta_{CA} & s_1 \Delta_{BC}^* & -\epsilon \end{pmatrix}. \quad (1)$$

The $3Q$ order parameter $(\Delta_{AB}, \Delta_{BC}, \Delta_{CA}) = \Delta(e^{i\phi_1}, e^{i\phi_2}, e^{i\phi_3})$ is classified by the total phase $\Phi = \phi_1 + \phi_2 + \phi_3 \bmod 2\pi \in \{0, \pi\}$. The central claim is that *phase frustration* among the ϕ_i , activated by the vH1–vH2 mixing λ , stabilizes the nematic phase NLCBO with $(\phi_1, \phi_2, \phi_3) = (0, \pi/2, \pi/2)$, which breaks both $C_6 \rightarrow C_2$ and time-reversal symmetry.

2 Method & provenance

We adapted the shared kagome loop-current kernel (`loop_current_kagome_kernel.py`, built for Fernandes *et al.* arXiv:2502.16657): reused its hexagonal-BZ / M -point geometry conventions and its “loop current = $\text{Im}\langle c_i^\dagger c_j \rangle$, charge = Re ” bond-operator philosophy. The paper’s object is not the plain 3×3 NN kagome model of the kernel but a 6×6 effective patch Hamiltonian, which we implemented directly (`code/patch_model.py`) from Eqs. (1),(4),(9),(11),(12). All physical constants are taken from the paper: $\epsilon = 0.12$ eV, $s_1 = -1.62$, $s_2 = 0.5$ (Fig. 5 caption), $\Delta = 0.2$ eV (Fig. 4), $\lambda \approx 0.35$ eV $\cdot a$, $k_{\text{cut}} = 1$. No parameter was fitted or invented.

3 Machine-checkable claims and results

ID	Claim	Result
C1	Numeric $H(\mathbf{k})$ at $\lambda = 0$ = analytic Eq. (9); degeneracy at $\Phi = 0, \pi$	PASS
C2	Φ /TRSB classification of orders; NLCBO uniquely nematic	PASS
C3	$1/\Delta E_1 > 0$, $1/\Delta E_2 < 0$ at $\Delta = 0.2$ (Eq. 12, Fig. 5)	PASS
C4	LCBO+ lowest at full fill; NLCBO anomalous k_x -dispersion (mechanism)	PASS
C5	λ required: $\Phi = \pi$ phases degenerate at $\lambda = 0$, split at $O(\lambda^2)$	PASS

C1. The 6×6 numeric eigenvalues at $\mathbf{k} = 0$, $\lambda = 0$ match the analytic $E_n^{(i)} = (-1)^i \epsilon - \mu + 2s_i \Delta \cos((\Phi + 2\pi n)/3)$ to $\max |\text{num} - \text{anal}| \approx 2.5 \times 10^{-16}$. At $\Phi = 0$ and $\Phi = \pi$ the spectrum collapses to 4 distinct levels (a 2+1 degeneracy in each block), whereas a generic $\Phi = 1.5$ gives 6 distinct levels—confirming the special degeneracy of Fig. 4.

C2. CBO+/LCBO− have $\Phi = 0$; CBO−/LCBO+/NLCBO have $\Phi = \pi$ (all reproduced). CBO $\pm$ are purely real (no loop current, TRSB=False); LCBO $\pm$ /NLCBO carry $\text{Im } \Delta \neq 0$ (TRSB=True). Only NLCBO has unequal phases $(0, \pi/2, \pi/2)$, i.e. it is the unique nematic (C_3 -breaking) configuration among these orders.

C3. Using Eq. (12) with $\epsilon = 0.12$, $s_1 = -1.62$, $s_2 = 0.5$: at $\Delta = 0.2$ we obtain $1/\Delta E_1 = +3.85 > 0$ and $1/\Delta E_2 = -4.39 < 0$, exactly the sign structure the paper requires (Fig. 5). The scan shows the crossover: at tiny $\Delta = 0.05$ both are positive; $1/\Delta E_2$ turns negative by $\Delta \approx 0.1$.

C4. Integrating Eq. (11) over a fully occupied patch, LCBO+ has the lowest correction (+763 vs. NLCBO +1101 vs. CBO− +3803, arb. units), matching the paper’s stated “if the band is fully occupied, LCBO+ has the lowest free energy.” The mechanism for NLCBO is the *anomalous dispersion*: along k_x the NLCBO band correction (−0.071 at the zone edge) is the most negative of the three (LCBO+ +0.034, CBO− +0.168), because its $+\frac{8}{3}k_x^2/\Delta E_2$ term with $1/\Delta E_2 < 0$ lowers the band; NLCBO is anisotropic ($k_x \neq k_y$) while CBO− is isotropic. This is precisely the ingredient that, under partial filling (condition iii), lets NLCBO win.

C5. The perturbative corrections of CBO−/LCBO+/NLCBO all vanish at $\lambda = 0$; the numeric filled-band internal energies of the three $\Phi = \pi$ configs coincide to $< 10^{-6}$ at $\lambda = 0$ and split ($\sim 4 \times 10^{-2}$) at $\lambda = 0.35$. Thus the degeneracy is lifted only by the vH1–vH2 mixing, exactly as claimed.

4 Verdict

Reproduced (analytic patch-model core). The paper’s central mechanism— phase-frustrated $3Q$ order, Φ classification, the λ -driven lifting of a $\Phi = \pi$ degeneracy, and the anomalous k_x dispersion that singles out the nematic NLCBO—is quantitatively reproduced with the paper’s own parameters and no free knobs.

Verdict: REPRODUCED (analytic core).

Coverage: 6/10 — full analytic patch model; *not* the mean-field simulated-annealing phase diagram (Figs. 2,3,7) or the 9-band DFT tight-binding model (Sec. IV, Fig. 6).

Agreement: 9/10 — every reproduced claim matches to machine precision or matches the paper’s stated qualitative outcome (signs, orderings, degeneracies).

5 What was not reproduced (honest scope)

The full mean-field phase diagrams (Figs. 2,3,7,9) require a self-consistent simulated-annealing minimization of the free energy over the 6-dimensional ($\Delta_{\alpha\beta}$) space on a ~ 500 -point (or 30×30) k -grid at $T = 90$ K, and the 9-band model requires DFT-derived hoppings from Ref. [74] that are not published in-line. These are multi-hour fitting/optimization tasks outside an overnight analytic replication and are documented in `failure_analysis.md`. The pieces we *did* reproduce are the model-defining equations and the perturbative mechanism, which are the load-bearing theoretical content.